

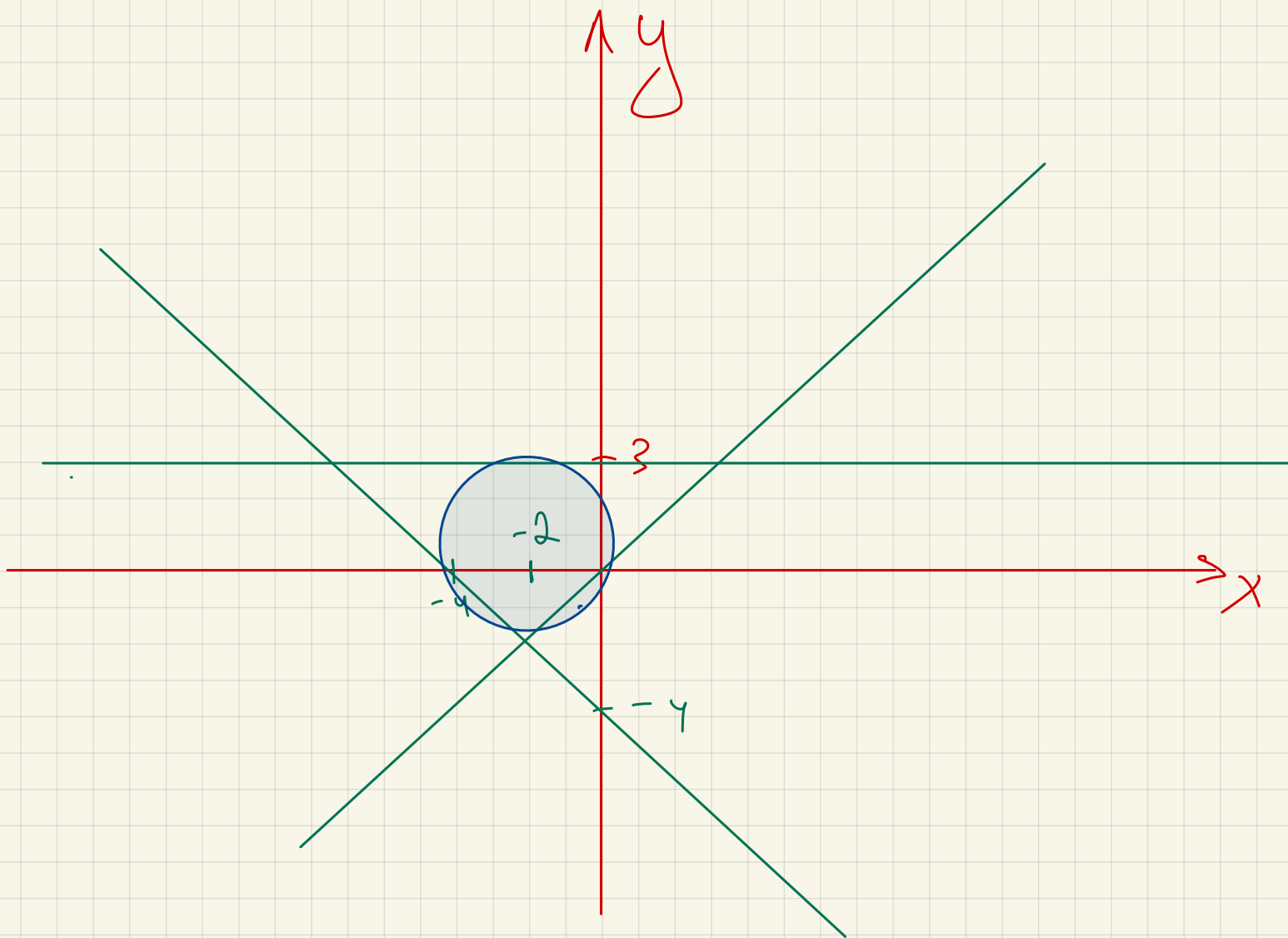


$$\textcircled{-} \frac{t^2 + 7t - 10}{(t-1)(t-3)} \geq 0$$

$$-\frac{(t-5)(t-2)}{(t-1)(t-3)} \geq 0$$

$$\begin{cases} (y-3)(y-x+4)(y-x) = 0 \\ (x+2)^2 + (y+3a)^2 = \underbrace{8a^2 + 24a + 4}_{R = \sqrt{8a^2 + 24a + 4}} \end{cases}$$

$(-2; -3a)$



$$x^2 - 3xy + 4y^2$$

$$x^2 - \frac{3xy}{2} + \frac{9y^2}{4} - \frac{9y^2}{4} + 4y^2$$

$$\left(x - \frac{3y}{2}\right)^2 - \frac{9y^2}{4} + 4y^2 = 0$$

$$4\left(x - \frac{3y}{2}\right)^2 - 9y^2 + 16y^2 = 0$$

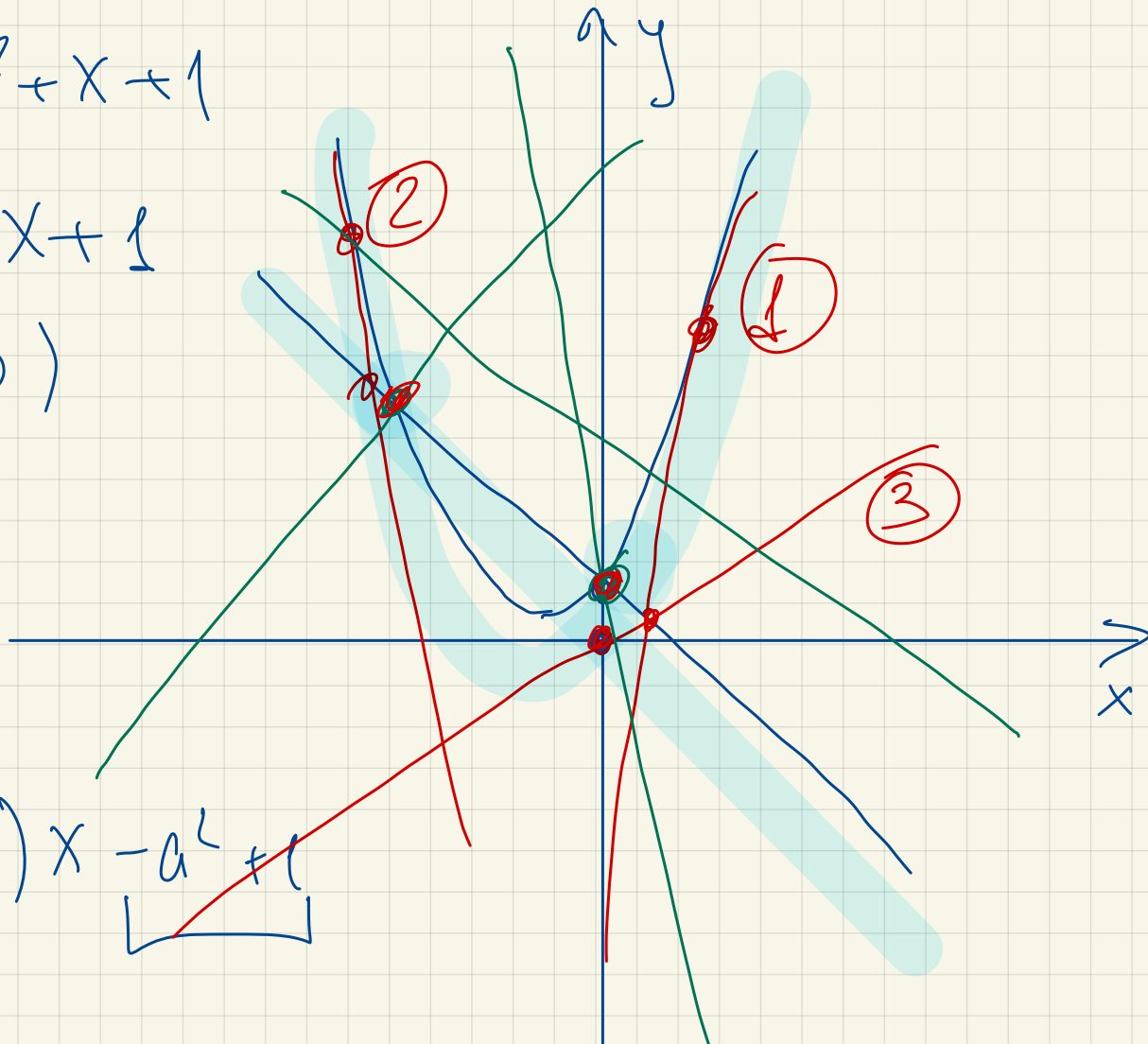
$$\boxed{4\left(x - \frac{3y}{2}\right)^2 + 7y^2 = 0} \quad (0;0)$$

$$x - \frac{3y}{2} = 0 \quad \Rightarrow \quad x = 0$$

$$7y^2 = 0 \quad \rightarrow \quad y = 0$$

$$\begin{cases} y = x^2 + x + 1 \\ y = -x + 1 \\ (0;0) \end{cases}$$

$$y = (2a+1)x - a^2 + 1$$



$$4a^2 - 4a^2$$

$$x^2 + x + 1 = (2a+1)x - a^2 + 1$$

$$x^2 + x - 2ax - 1 + a^2 = 0$$

$$(x-a)^2 = 0$$

$$x = a$$

$$(a; a^2 + a + 1)$$

т. касания

$$a = \pm 1 \leftarrow$$

$$a = -1$$

$$(2a+1)x - a^2 + 1 = -x + 1$$

$$x = ?$$

$$\left(\frac{a^2}{2a+1}, \frac{-1-a^2+2a+2}{2a+2} \right)$$

т. пересечения

$$-a^2 + 2ax + 2x = 0$$

$$x(2a+2) = a^2$$

$$x = \frac{a^2}{2a+2} \leftarrow$$

$$y = \frac{-a^2}{2a+2} + 1$$

$$a = 0$$

$$a = -2$$

$$a = 1$$

X

→

1 т. с 2р

→

1 т. с 2р

→

3 т. с 2р

ОСлабеем все $\log x$

$$a \in (-\infty; -2) \cup (-2; 0) \cup (0; 1) \cup (1; +\infty)$$

$$\begin{cases} (|x-a| + 2|y-1| - 4) (x^2 + (y-1)^2 - 1) \cdot \ln(y-x^2) = 0 \\ (|x-a| + 2|y-1| - 4) \left(\frac{|a|}{15} (x-1) - y + 5 \right) = 0 \end{cases}$$

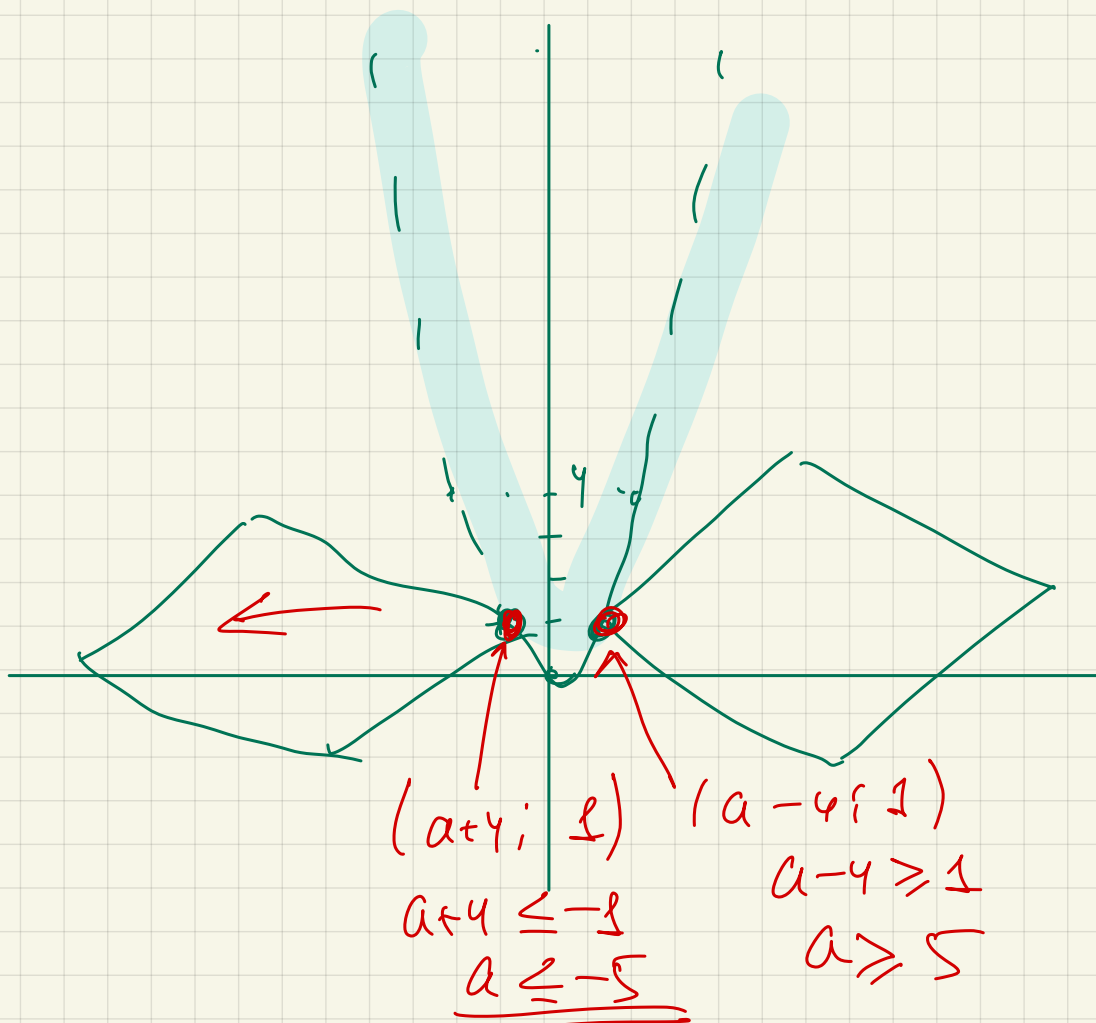
4 псу

$$\ln(y-x^2)$$

$$y-x^2 > 0$$

$$y > x^2$$

$$x^2 + (y-1)^2 = 1$$

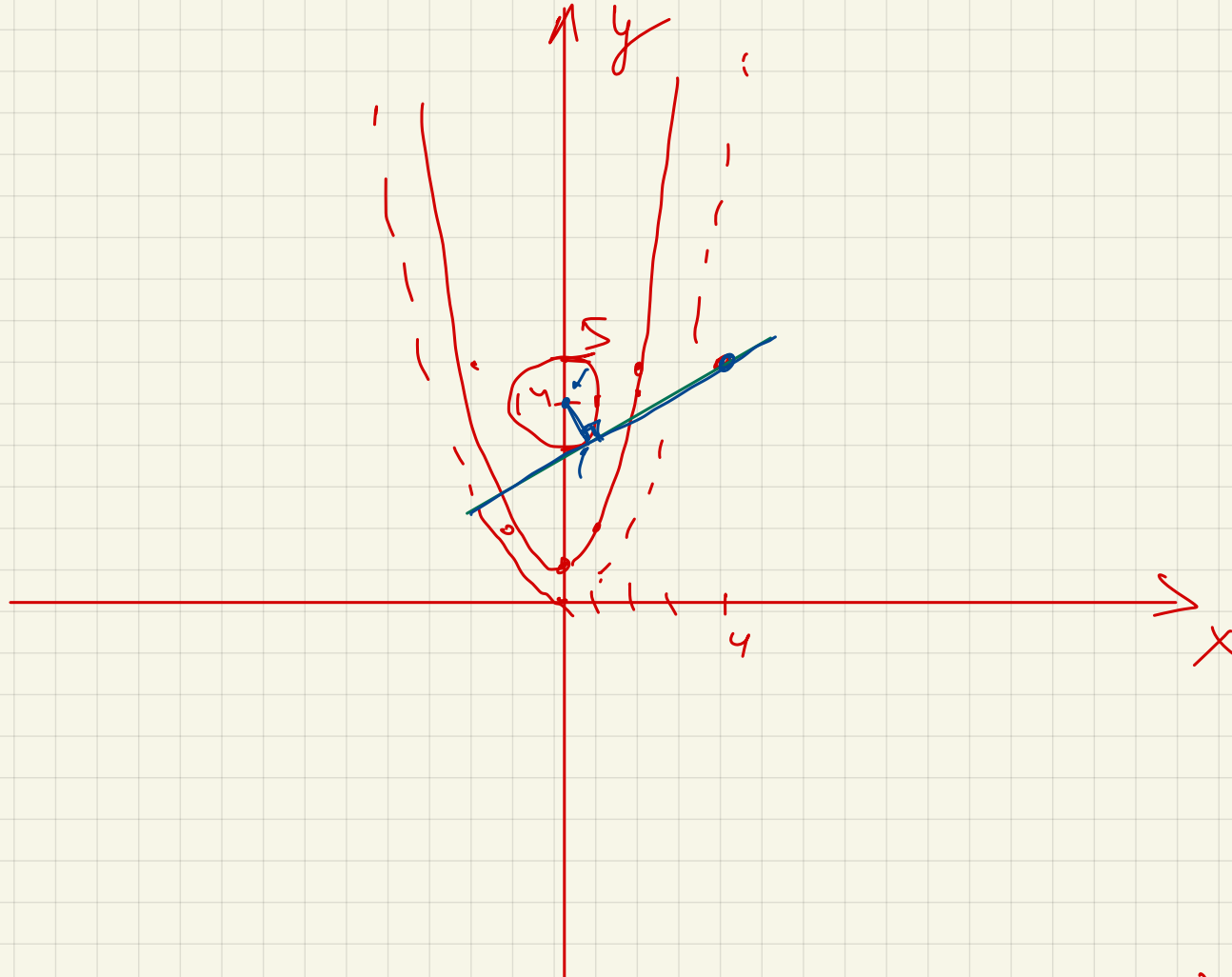


$$a \in (-\infty; -5) \cup (5; +\infty)$$

→ correct
even
per

$$\left\{ \begin{array}{l} (x^2 + (y-4)^2 - 1) \ln(y-x^2) = 0 \\ \frac{|a|}{15} (x-4) - y + 5 = 0 \\ \underline{a \in (-\infty; -5) \cup (5; +\infty)} \end{array} \right.$$

$$\begin{aligned} y - x^2 &= 1 \\ y &= x^2 + 1 \end{aligned}$$



$$y = \frac{|a|}{15} (x-4) + 5 \quad (4; 5)$$

$$x^2 + \left(\frac{|a|}{15} (x-4) + 5 - 4 \right)^2 = 1$$

$$x^2 + \left(\frac{|a|(x-4)}{15} + 1 \right)^2 = 1$$

$$x^2 + \left(\frac{|a|(x-4)}{15} \right)^2 + \frac{2 \cdot |a|(x-4)}{15} + x - x = 0$$

$$x^2 + \frac{a^2(x-4)^2}{15^2} + 2 \frac{|a|(x-4)}{15} = 0$$

$$x^2 + \frac{a^2(x^2 - 8x + 16)}{225} + \frac{2|a|(x-4)}{15} = 0$$

$$x^2 \left(1 + \frac{a^2}{225}\right) - x \left(\frac{8a^2}{225} + \frac{2|a|}{15}\right) + \frac{16a^2}{225} - \frac{8|a|}{15} = 0$$

$$\left(\frac{8a^2}{225} + \frac{2|a|}{15}\right)^2 - 4 \cdot \left(1 + \frac{a^2}{225}\right) \left(\frac{16a^2}{225} - \frac{8|a|}{15}\right) = 0$$

$$\frac{64a^4}{15^4} + \frac{2 \cdot |a| \cdot 8a^2}{15^2} \cdot 2 + \frac{4a^2}{225}$$

$$- 4 \left(\frac{16a^2}{225} - \frac{8|a|}{15} + \frac{16a^4}{15^4} - \frac{8|a|a^2}{15^3} \right)$$

$$\frac{64a^4}{15^4} + \frac{2|a| \cdot 8a^2 \cdot 2}{15^3} + \frac{4a^2}{15^2} - \frac{64a^2}{15^2}$$

$$+ \frac{32|a|}{15} - \frac{64a^4}{15^4} + \frac{32|a|a^2}{15^3}$$

$$- \frac{60a^2}{15^2} + \frac{32|a|}{15} + \frac{64|a|a^2}{15^3} = 0$$

$$\cancel{\frac{32|a|}{15}} + \frac{64|a|a^2}{15^3} - \frac{60a^2}{15^2} = 0$$

$$a = 0; 8; -8$$